Machine Learning for Spotify Genre Classification

Group 8:

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1) Problem Statement

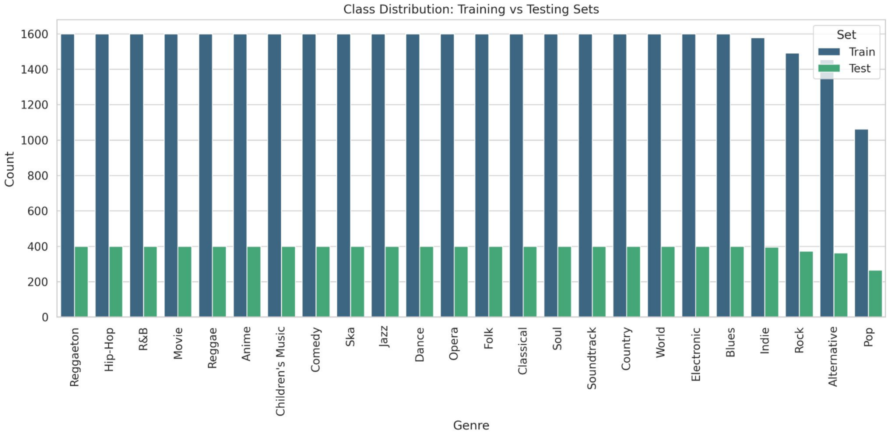
- **The Goal:** Predict the musical genre of a track (1 of 24 classes) based on audio features and metadata.
- **The Challenge:**
  - **Subjectivity:** Genre boundaries are fluid (e.g., "Indie" vs. "Alternative").
  - **Data Format:** We are using tabular summary data, not raw audio waveforms.
  - **High Cardinality:** 24 distinct classes makes this a difficult multi-class problem.
- **Relevance:** Useful for recommendation engines (Spotify/Apple Music) and playlist generation.

2) Dataset Collection & Description

- **Source:** Kaggle (pulled from Spotify Web API).
- **Volume:** ~47,000 tracks (Post-cleaning).
- **Target Variable:** Genre (24 classes: Pop, Rock, Hip-Hop, Classical, Anime, etc.).
- **Input Features (14 Total):**
    - **Audio Metrics:** Acousticness, Danceability, Energy, Instrumentalness, Liveness, Loudness, Speechiness, Tempo, Valence.
    - **Metadata:** Popularity, Duration.
    - **Structure:** Key, Mode, Time Signature.

3) Data Preprocessing Techniques

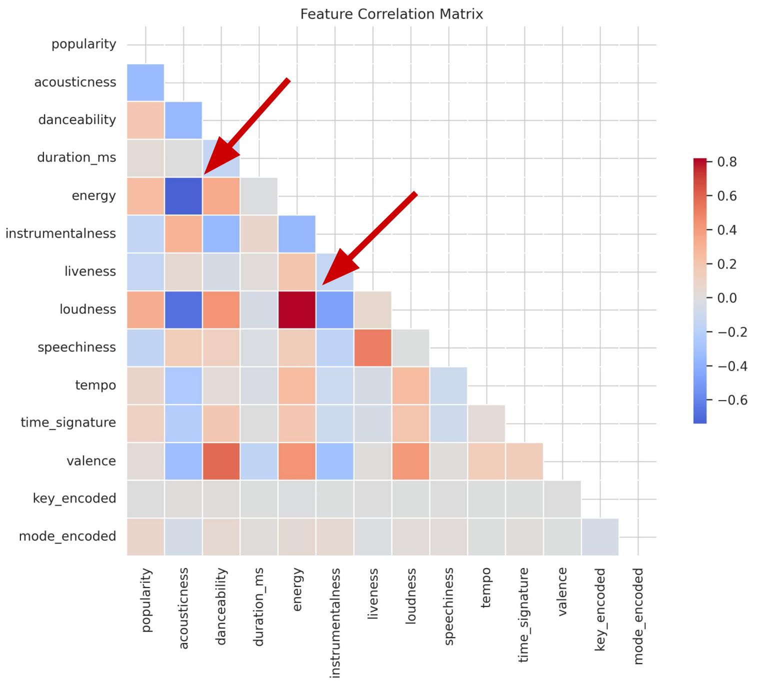
- **Data Cleaning:**
  - Removed tracks with multiple genre labels to ensure single-label classification.
  - Merged "Rap" into "Hip-Hop" due to high feature overlap.
  - Dropped "A Capella" due to insufficient sample size.
- **Hybrid Balancing Strategy:**
  - **Issue:** Severe class imbalance in raw data.
  - **Solution:** Undersampled majority classes to a cap of 2,000 samples; kept minority classes intact.
- **Scaling & Encoding:**
  - StandardScaler applied to continuous features (crucial for KNN and MLP).
  - LabelEncoder applied to categorical features (Key, Mode).



Genre distribution after preprocessing

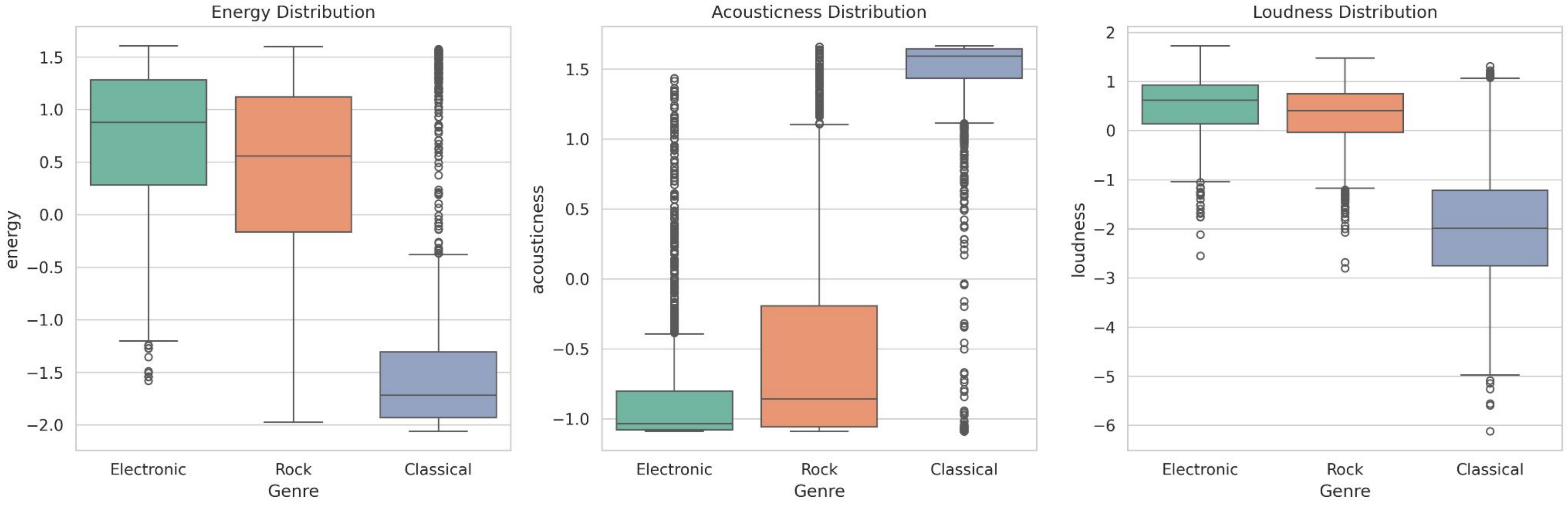
Data Examination

- **Feature Correlations:**
  - Strong Positive Correlation (r=0.82): **Energy vs. Loudness**.
  - Strong Negative Correlation (r=−0.74): **Acousticness vs. Energy**.
- **Feature Separability:**
  - *Example: Acousticness and Loudness* help separate **Classical** from **Electronic** and **Rock**.
  - *However*, each feature taken on its own does not have enough information for robust separation.



Feature correlation matrix

Data Examination (Cont.)



Key features show clear differences across genres, indicating their predictive value for classification.

5/6) Modeling Techniques Chosen

We evaluated 6 distinct models to cover a variety of mathematical approaches:

1. **Logistic Regression:** Good linear baseline
2. **K-Nearest Neighbors (KNN):** Distance-based
3. **Gaussian Naive Bayes:** Probabilistic
4. **Random Forest:** Bagging Ensemble
5. **XGBoost:** Boosting Ensemble
6. **Multilayer Perceptron (MLP):** Deep Learning

Initially, we considered a Support Vector Classifier (SVC), but we dropped it due to computational considerations.

7) Feature Selection

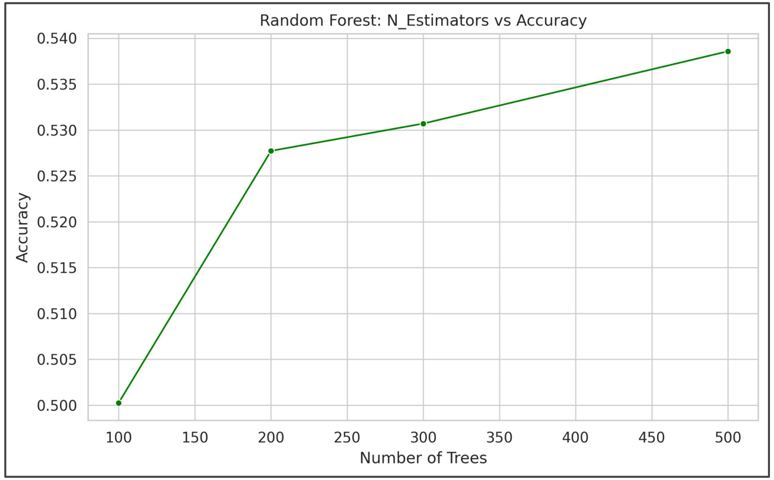
- **Manual Removal:**
  - Dropped artist_name, track_name, track_id.
  - ***Reasoning***: These are high-cardinality identifiers. We want the model to learn *acoustic characteristics*, not memorize specific artists.
- **Handling Multicollinearity:**
  - Retained correlated features (Energy/Loudness).
  - ***Reasoning***: Tree-based models (XGBoost, Random Forest) handle collinearity well via split selection. Deep Learning (MLP) can capture nonlinear interaction effects between them.
- **Final Set:** 14 Features utilized.

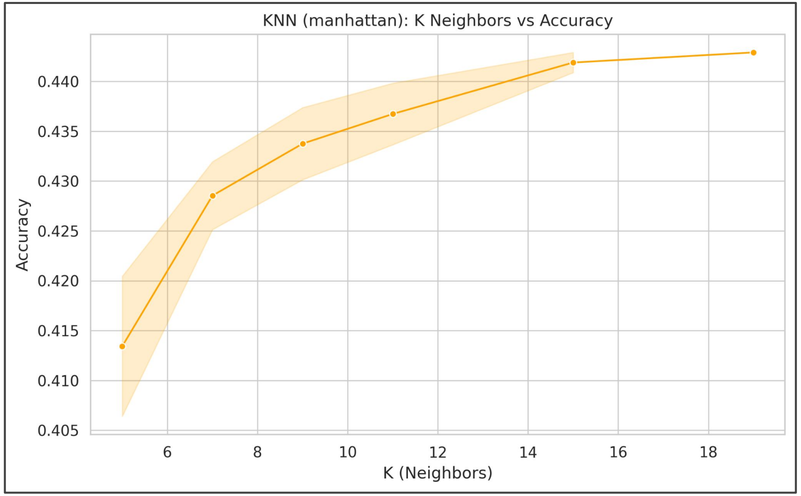
Hyperparameter Tuning

Method: Grid Search, with some models using randomized search.

Key Optimizations:

- **KNN:** Optimal k=19 with Manhattan distance
- **XGBoost:** Learning rate = 0.1, max depth 6.
- **Random Forest:** Stabilized at 500 estimators.
- **MLP:** 4-layer architecture, AdamW optimizer, Dropout = 0.3.





Logistic Regression/Classifier

Methodology:

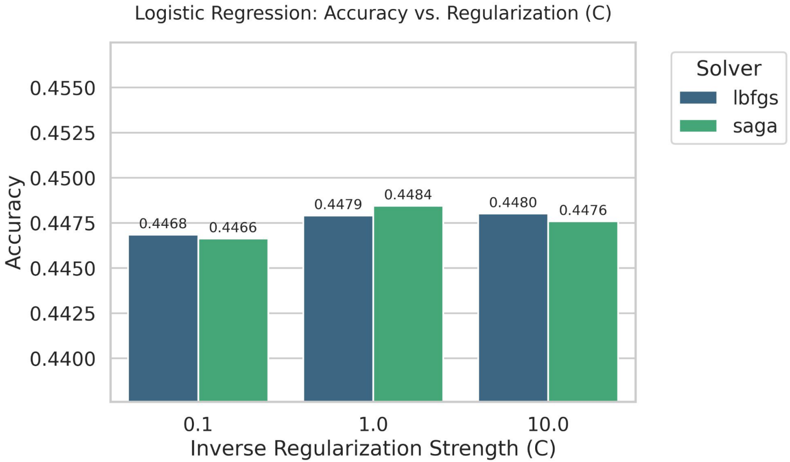
- Goal: Establish a baseline for linear separability.
- Tuning: Exhaustive Grid Search on Regularization strength (C) and Solvers (lbfgs, saga).

Statistical Analysis:

- Best Accuracy: 44.84% (with C=1.0).
- Std Dev across trials: 0.0007 (Extremely Low).
- Correlation: Parameter C had a weak positive correlation (r = 0.33).

Key Insight:

- The extremely low standard deviation across all trials indicates the model hit a hard ceiling.
- Changing solvers or penalties had almost no effect, proving that genre boundaries in this feature space are fundamentally non-linear.



KNN

Methodology:

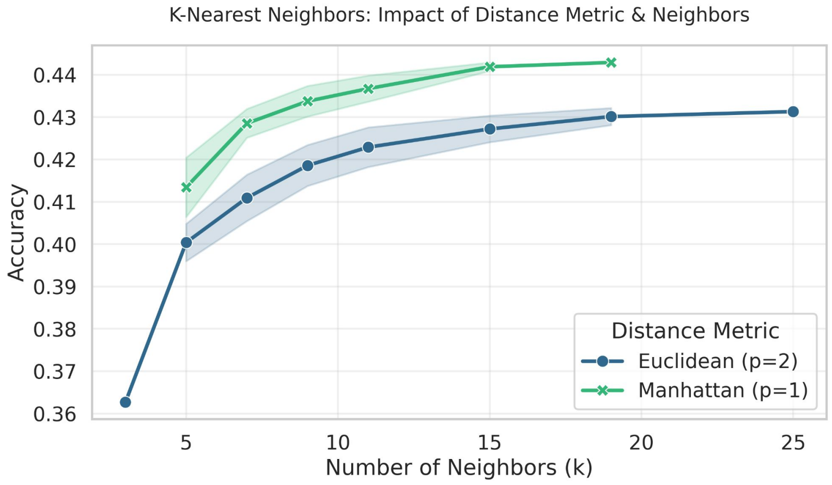
- Goal: Test if genres form distinct geometric clusters.
- Tuning: Neighbors (k), Weights, and Distance Metric. Exhaustive Grid Search.

Statistical Analysis:

- Exhaustive Search: 44.29% (with k=19).
- Grid Search: 45.46% (with k=25)
- Strong positive correlation between k and accuracy (r = 0.62).
- Distance Metric: Negative correlation with accuracy (r = -0.47).

Key Insight:

- Manhattan Distance (p=1) outperformed Euclidean (p=2).
- In high-dimensional audio space, the Manhattan (block) distance proved more robust to noise than standard straight-line distance.



Gaussian NB

Methodology:

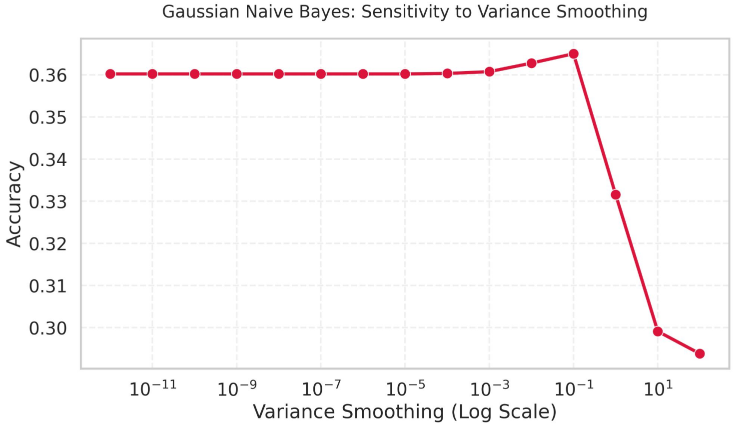
- Goal: Test a probabilistic approach assuming feature independence.
- Tuning: Exhaustive Grid search of var_smoothing.

Statistical Analysis:

- Best Accuracy: 36.50% (Lowest of all models).
- Key Correlation: Strong negative correlation with smoothing (r = -0.74).

Key Insight:

- The Independence Assumption Failed.
- Our EDA showed Energy and Loudness had a correlation of 0.82.
- Because the features were not independent, the model failed to capture the signal, performing significantly worse than the linear baseline.



Random Forest

Methodology:

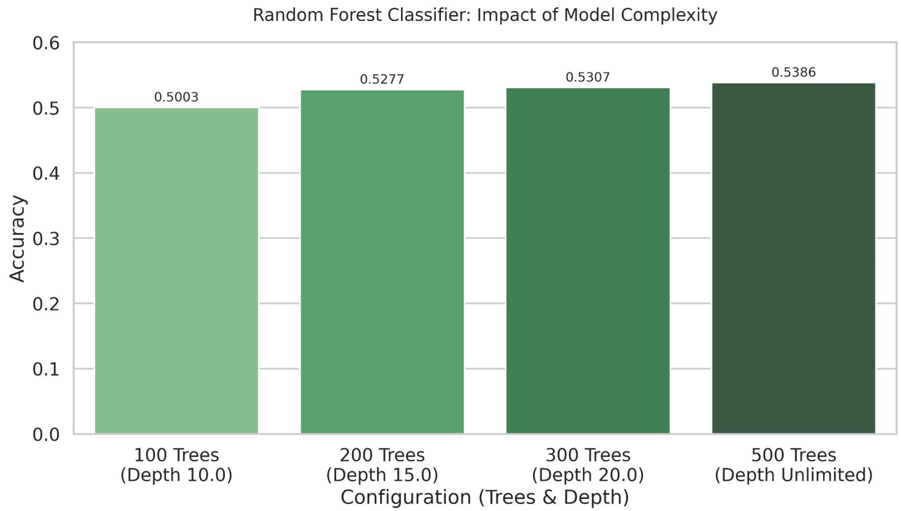
- Goal: Reduce variance through bagging (Bootstrap Aggregating).
- Params: n_estimators (Trees), max_depth.
- Tested 4 manual configurations.

Statistical Analysis:

- Best Accuracy: 53.86%.
- max_depth: r = 0.91 (Extremely High).
- n_estimators: r = 0.86 (Very High).

Key Insight:

- Unlike Logistic Regression, this model did not plateau.
- The massive correlation with depth (0.91) suggests that genre decision boundaries are highly complex and require deep, non-linear splits to classify correctly.



XGBoost

Methodology:

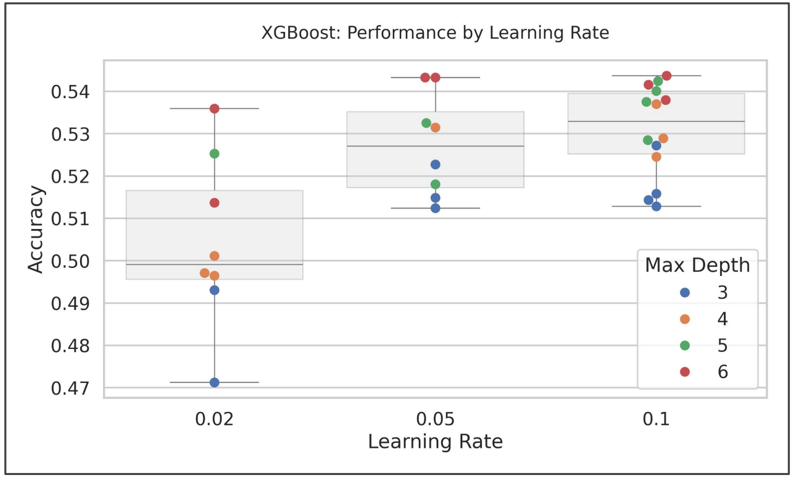
- Goal: Reduce bias through Gradient Boosting.
- Params: Max Depth, Estimators, LR...
- Tuning: 30 trials using Randomized Search

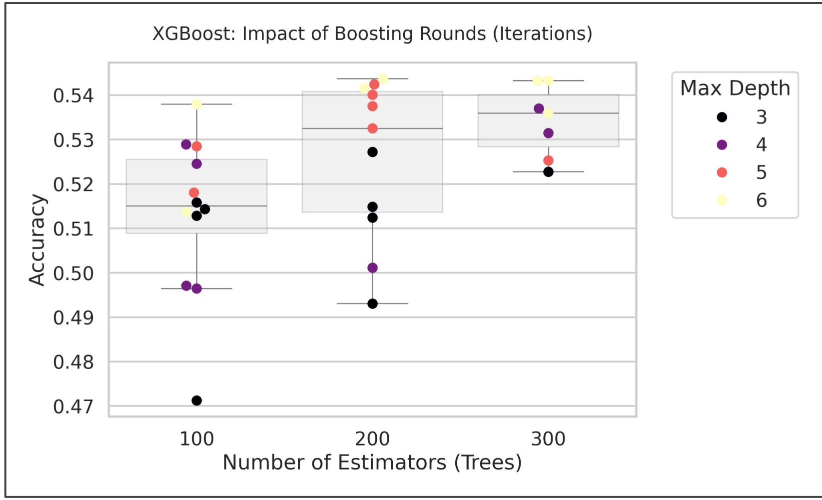
Statistical Analysis:

- Best Accuracy: 54.37% (The Winner).
- max_depth: r = 0.64 (Moderate positive).
- learning_rate: r = 0.56 (Moderate positive).

Key Insight:

- XGBoost achieved the highest single accuracy.
- Optimal configuration was a Learning Rate of 0.1 combined with a Depth of 6. Lower learning rates (0.02) consistently underperformed.





MLP (Neural Network)

Methodology:

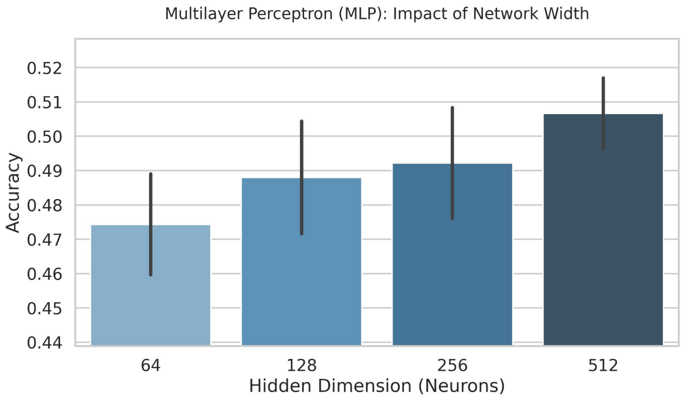
- Architecture: Custom PyTorch Feed-Forward Network.
- Params: layer width, depth, learning rate, batch size.
- Search Method: Randomized Grid Search

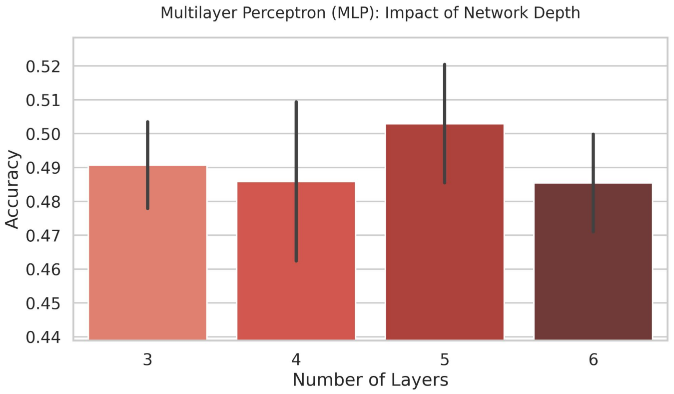
Statistical Analysis:

- Best Accuracy: 51.07%.
- Architecture Correlations:
- Width (hidden_dim): r = 0.60 (Strong Positive).
- Depth (num_layers): r = -0.02 (No Correlation).

Key Insight:

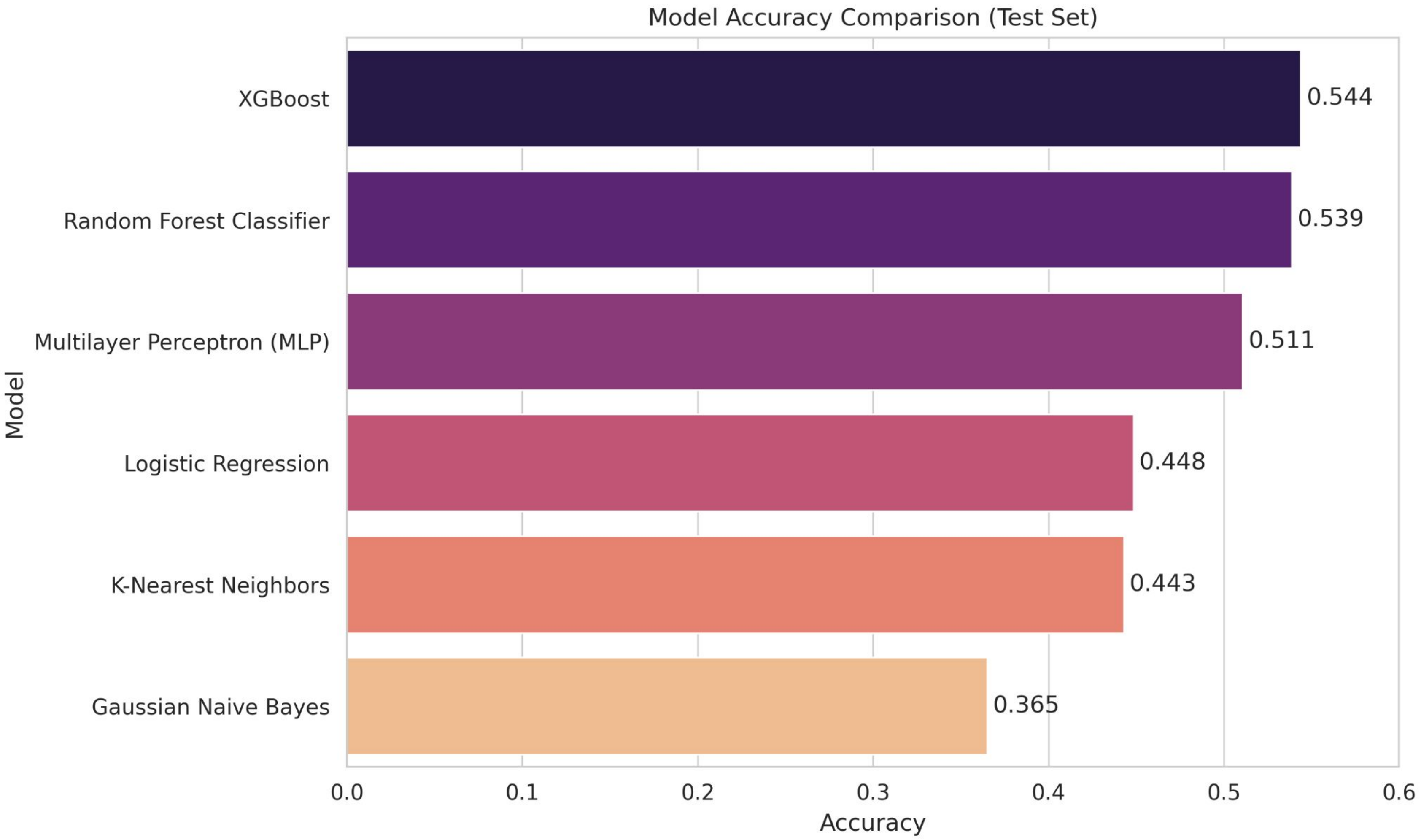
- Width is greater than Depth: For this tabular dataset, a wider network (512 neurons) was more effective than a deeper network.
- Adding more layers (depth) added computational cos without improving accuracy.





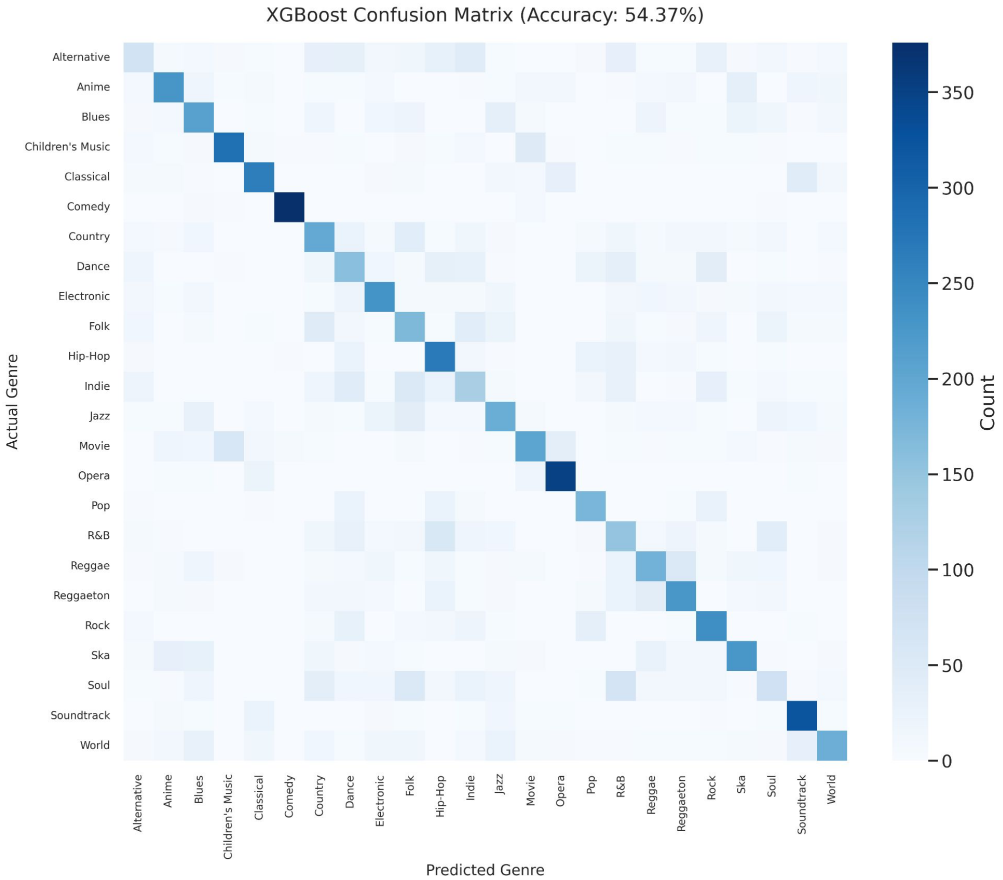
8) Results

Model	Accuracy	Weighted F1
XGBoost	**54.37%**	**0.5400**
Random Forest	53.86%	0.5322
MLP (Neural Net)	51.07%	0.5046
Logistic Regression	44.84%	0.4369
KNN	44.29%	0.4402
Gaussian NB	36.50%	0.3405
Random Classifier	4.17%	0.0417



Model Accuracy Comparison (Test Set)
XGBoost
0.544
Random Forest Classifier
0.539
Multilayer Perceptron (MLP)
0.511
Logistic Regression
0.448
K-Nearest Neighbors
0.443
Gaussian Naive Bayes
0.365
Model
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Accuracy

Confusion Matrix (XGB, Best Trial)



9) Interpretation

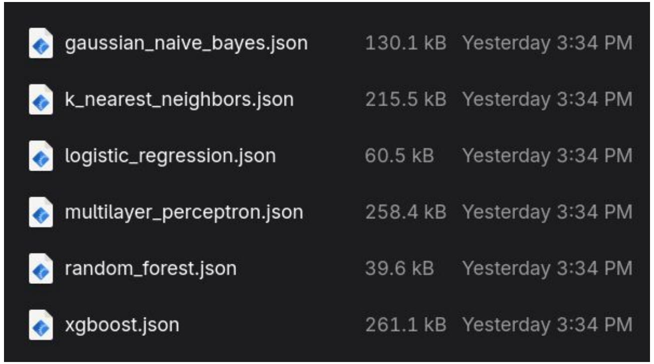
1. **Tree Ensembles Dominate:**
   - Genre boundaries are "ragged" and non-linear. Decision trees partition this space better than hyperplanes (Logistic Regression) or clusters (KNN).
2. **Impact of Correlation:**
   - Naive Bayes failed (36.5%) likely because it assumes features are independent. The strong correlation between Energy/Loudness violated this assumption.
3. **The 55% Ceiling:**
   - Why not higher?
   - **Subjectivity:** Human labeling is inconsistent.
   - **Data Limitation:** Tabular features lose temporal information (rhythm variations, lyrical content, timbre changes).

9) Recommendations

1. **Raw Audio Processing:**
   - Move away from more basic audio features. Use **Convolutional Neural Networks (CNNs)** or a pre-trained audio embedding model like **CLAP**.
2. **Hierarchical Classification:**
   - Predict "Super-Genres" first (e.g., Rock) then predict Sub-Genres (e.g., Alternative Rock).
3. **Dimensionality Reduction:**
   - Apply PCA for distance-based models (KNN) to reduce noise from correlated features.

10) Collaboration

- Modular setup to streamline the project.
- One preprocessing pipeline ensured all models used identical Train/Test sets.
- Each member handled one model and its tuning.
- Results exported to strict JSON schema (Hyperparams, Confusion Matrix).
- Shared script aggregated JSONs to evaluate models and generate plots programmatically.



Conclusion

- **Summary:**
  - Successfully processed 47k tracks across 24 genres.
  - Addressed class imbalance and multicollinearity.
  - **XGBoost** identified as the optimal model (54.37% Accuracy).
- **Key Takeaway:**
  - Audio summary features contain significant predictive signal (13x better than random guessing), but are insufficient for human-level classification accuracy.
- **Code Availability:**
  - https://github.com/carterprince/spotify-classification